

Part 4 — Perform a Simple Length-Filtering Experiment

Metric	Original genome	After ≥ 10 kb filter
Total length	2,423,263,331	2,422,532,368
Number of sequences	481	371
Maximum length	108,574,468	108,574,468
N50	66,396,233	66,396,233
GC content	41.56%	41.56%

- The table shows that applying the 10 kb filter removed 110 short sequences, reducing the total genome length slightly. However, the maximum sequence length and N50 remained the same. The GC content also stayed at 41.56%. This means that although some short sequences were removed, the main characteristics of the genome assembly were not greatly affected.